

1 Summary

Model (Run-name): model3 (test)

Iteration: 16

TAC: 64130.517426821825

Took: 3413.94774889946s

Total Time: 9950.238433599472s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

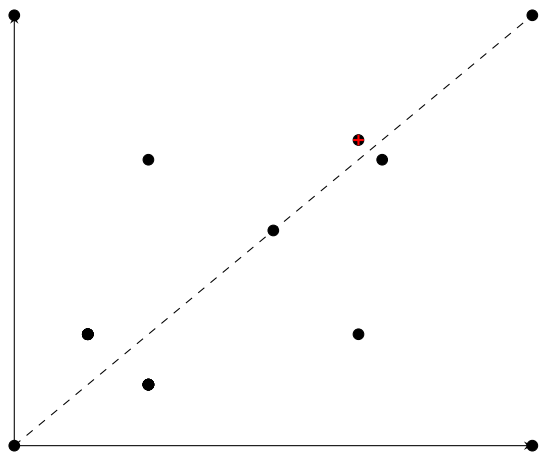
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{3,5,1}$	22.3	$q_{3,5,1}$	1544.29000000000002
$z_{2,1,1}$	1.0	$dt_{3,5,2}$	29.056081081080922	$q_{4,3,1}$	1539.72
$z_{2,2,1}$	1.0	$dt_{4,3,1}$	50.0	$q_{5,4,1}$	1634.85
$z_{3,5,1}$	1.0	$dt_{4,3,2}$	111.1	$q_{cu,1}$	357.40999999997564
$z_{4,3,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{cu,3}$	839.9899999999769
$z_{5,4,1}$	1.0	$dt_{5,4,2}$	24.33559322033887	$q_{cu,5}$	724.5600000000245
$z_{cu,1}$	1.0	$dt_{cu,1}$	93.91477272726996	$t_{1,1}^{(H)}$	160.0
$z_{cu,3}$	1.0	$dt_{cu,3}$	82.35608108107952	$t_{1,2}^{(H)}$	160.0
$z_{cu,5}$	1.0	$dt_{cu,5}$	66.53559322034036	$t_{1,3}^{(H)}$	133.91477272726996
$A_{1,1,2}^{\beta}$	2.177072042101825	$f_{1,1,2}^H$	3.441529235382455	$t_{2,1}^{(H)}$	248.9
$A_{2,1,1}^{\beta}$	3.8456457514706726	$f_{2,1,1}^H$	4.774527452745182	$t_{2,2}^{(H)}$	137.8
$A_{2,2,1}^{\beta}$	7.838821872890362	$f_{2,2,1}^H$	5.825472547254725	$t_{2,3}^{(H)}$	137.8
$A_{3,5,1}^{\beta}$	12.923002798326602	$f_{3,5,1}^H$	9.585909373060211	$t_{3,1}^{(H)}$	226.7
$A_{4,3,1}^{\beta}$	6.676083254111166	$f_{4,3,1}^H$	12.6	$t_{3,2}^{(H)}$	122.35608108108028
$A_{5,4,1}^{\beta}$	14.143774763563352	$f_{5,4,1}^H$	12.264441110277563	$t_{3,3}^{(H)}$	122.35608108107952
$A_{cu,1}^{\beta}$	2.682203014960301	$f_{1,1,2}^C$	2.581871104206264	$t_{4,1}^{(H)}$	271.1
$A_{cu,3}^{\beta}$	5.422783698740899	$f_{2,1,1}^C$	7.6	$t_{4,2}^{(H)}$	148.9
$A_{cu,5}^{\beta}$	5.324293809465329	$f_{2,2,1}^C$	6.1	$t_{4,3}^{(H)}$	148.9
$A_{1,1,2}$	3.7593939060848847	$f_{3,5,1}^C$	13.9	$t_{5,1}^{(H)}$	198.9
$A_{2,1,1}$	9.439495705509202	$f_{4,3,1}^C$	8.4	$t_{5,2}^{(H)}$	106.535593220339
$A_{2,2,1}$	30.932717773740634	$f_{5,4,1}^C$	17.3	$t_{5,3}^{(H)}$	106.53559322034036
$A_{3,5,1}$	71.16597394333074	$RecLMTD_{1,1,2}$	0.013920648312654002	$t_{1,1}^{(C)}$	160.0
$A_{4,3,1}$	23.670259536214445	$RecLMTD_{2,1,1}$	0.01512597106171161	$t_{1,2}^{(C)}$	90.20394736842427
$A_{5,4,1}$	82.71968770372004	$RecLMTD_{2,2,1}$	0.04062485145112026	$t_{1,3}^{(C)}$	60.0
$A_{cu,1}$	5.22801241844764	$RecLMTD_{3,5,1}$	0.03917080202023656	$t_{2,1}^{(C)}$	221.7
$A_{cu,3}$	16.73801144031253	$RecLMTD_{4,3,1}$	0.013067129481842327	$t_{2,2}^{(C)}$	115.6
$A_{cu,5}$	16.23442672627479	$RecLMTD_{5,4,1}$	0.04300806468370923	$t_{2,3}^{(C)}$	115.6
$dt_{1,1,2}$	69.79605263157573	$RecLMTD_{cu,1}$	0.012433369395597542	$t_{3,1}^{(C)}$	221.1
$dt_{1,1,3}$	73.91477272726996	$RecLMTD_{cu,3}$	0.01693747511787797	$t_{3,2}^{(C)}$	37.8
$dt_{2,1,1}$	88.9	$RecLMTD_{cu,5}$	0.019045024176509287	$t_{3,3}^{(C)}$	37.8
$dt_{2,1,2}$	47.596052631575745	$q_{1,1,2}$	229.55	$t_{4,1}^{(C)}$	176.7
$dt_{2,2,1}$	27.2	$q_{2,1,1}$	530.45	$t_{4,2}^{(C)}$	82.2
$dt_{2,2,2}$	22.2	$q_{2,2,1}$	647.2099999999999	$t_{4,3}^{(C)}$	82.2

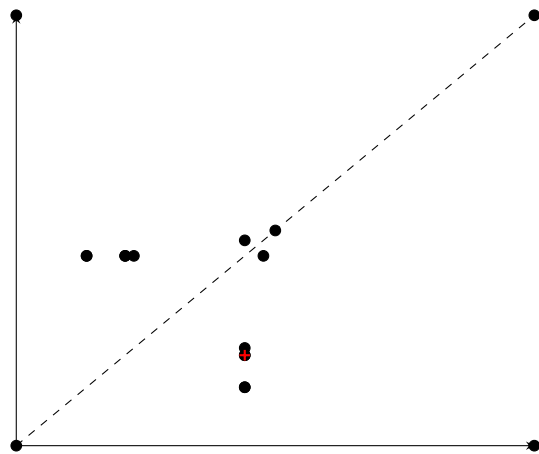
Variable	Value	Variable	Value
$t_{5,1}^{(C)}$	204.4	$b_{1,1,2}^{C,\text{out}}$	384.46226625240024
$t_{5,2}^{(C)}$	93.3	$b_{2,1,1}^{C,\text{out}}$	1216.0
$t_{5,3}^{(C)}$	93.3	$b_{2,2,1}^{C,\text{out}}$	1352.37
$t_{1,1,2}^H$	93.3	$b_{3,5,1}^{C,\text{out}}$	2841.16
$t_{2,1,1}^H$	138.04840635580382	$b_{4,3,1}^{C,\text{out}}$	1857.2400000000002
$t_{2,2,1}^H$	137.98823929830698	$b_{5,4,1}^{C,\text{out}}$	3056.91
$t_{3,5,1}^H$	65.6		
$t_{4,3,1}^H$	148.9		
$t_{5,4,1}^H$	84.37210141374759		
$t_{1,1,2}^C$	148.9083888138303		
$t_{2,1,1}^C$	160.0		
$t_{2,2,1}^C$	221.7		
$t_{3,5,1}^C$	204.4		
$t_{4,3,1}^C$	221.1		
$t_{5,4,1}^C$	176.7		
$b_{1,1,2}^{H,\text{in}}$	550.6446776611928		
$b_{2,1,1}^{H,\text{in}}$	1188.3798829882758		
$b_{2,2,1}^{H,\text{in}}$	1449.960117011701		
$b_{3,5,1}^{H,\text{in}}$	2173.1256548727497		
$b_{4,3,1}^{H,\text{in}}$	3415.8599999999997		
$b_{5,4,1}^{H,\text{in}}$	2439.3973368342076		
$b_{1,1,2}^{H,\text{out}}$	321.09467766116836		
$b_{2,1,1}^{H,\text{out}}$	657.9298829883		
$b_{2,2,1}^{H,\text{out}}$	802.750117011701		
$b_{3,5,1}^{H,\text{out}}$	628.8356548727497		
$b_{4,3,1}^{H,\text{out}}$	1876.1399999999999		
$b_{5,4,1}^{H,\text{out}}$	804.5473368342082		
$b_{1,1,2}^{C,\text{in}}$	154.91226625237584		
$b_{2,1,1}^{C,\text{in}}$	685.55		
$b_{2,2,1}^{C,\text{in}}$	705.16		
$b_{3,5,1}^{C,\text{in}}$	1296.8699999999997		
$b_{4,3,1}^{C,\text{in}}$	317.52		
$b_{5,4,1}^{C,\text{in}}$	1422.06		

3 *RecLMTD* Tangent Points

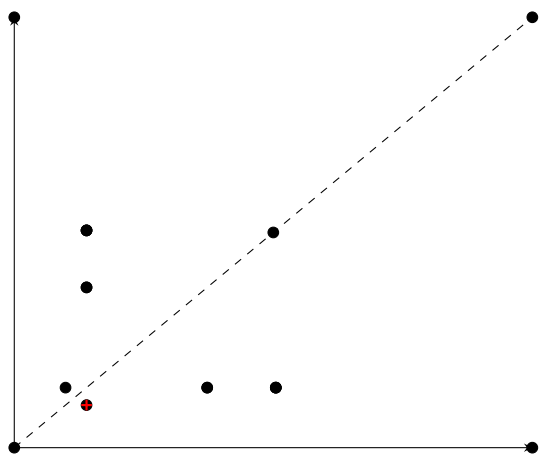
- (1, 1, 2)
 - Value: 0.013921
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000014
- (2, 1, 1)
 - Value: 0.015126
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000034
- (2, 2, 1)
 - Value: 0.040625
 - Error: 0.000000085433623
 - Relative error: 0.000002102984764
- (3, 5, 1)
 - Value: 0.039171
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000003
- (4, 3, 1)
 - Value: 0.013067
 - Error: 0.000000099506986
 - Relative error: 0.000007615002885
- (5, 4, 1)
 - Value: 0.043008
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000003
- $cu, 1$
 - Value: 0.012433
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000009
- $cu, 3$
 - Value: 0.016937
 - Error: 0.000000997394721
 - Relative error: 0.000058883392252
- $cu, 5$
 - Value: 0.019045
 - Error: 0.000000962937250
 - Relative error: 0.000050558537310



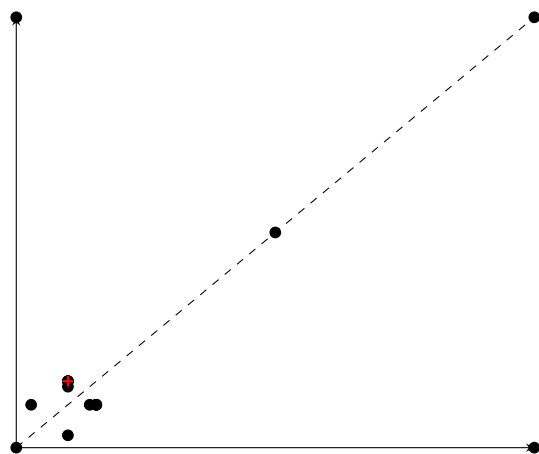
(a) RecLMTD - (1, 1, 2)



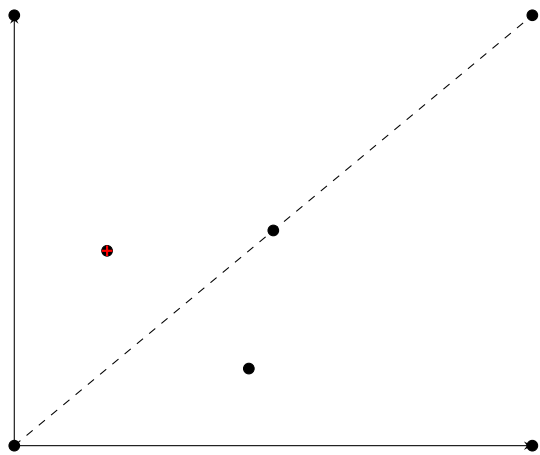
(b) RecLMTD - (2, 1, 1)



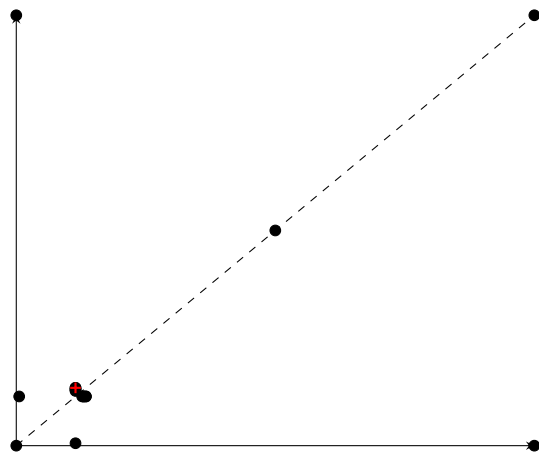
(c) RecLMTD - (2, 2, 1)



(d) RecLMTD - (3, 5, 1)



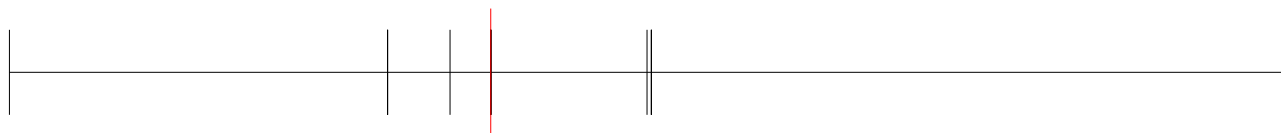
(e) RecLMTD - (4, 3, 1)



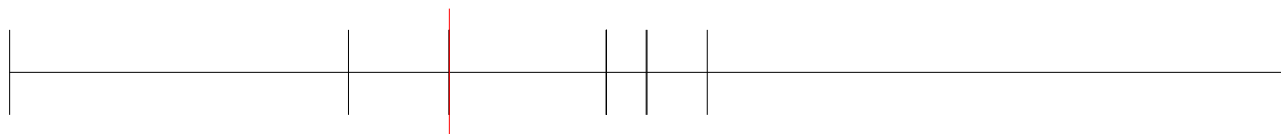
(f) RecLMTD - (5, 4, 1)



(a) RecLMTD - cu, 1



(b) *RecLMTD* - $cu, 3$



(c) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 550.6446776611928
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1188.3798829882758
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.960117011701
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2173.1256548727497
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 4, 1)
 - Value: 2439.3973368342076
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.8641790905479326e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 321.09467766116836
 - Absolute error: 1.4665602066088468e-11
 - Relative error: -4.5673762557856886e-14
- (2, 1, 1)
 - Value: 657.9298829883
 - Absolute error: 1.1860229652077123
 - Relative error: -0.0017994148745234062
- (2, 2, 1)

- Value: 802.750117011701
 - Absolute error: 1.096582864601828
 - Relative error: -0.00136416914415469
- (3, 5, 1)
 - Value: 628.8356548727497
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.807894270000029e-16
- (4, 3, 1)
 - Value: 1876.1399999999999
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.2119227533298796e-16
- (5, 4, 1)
 - Value: 804.5473368342082
 - Absolute error: 230.22933230506533
 - Relative error: -0.2224918083015632

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 154.91226625237584
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 685.55
 - Absolute error: 2.4442670110147446e-11
 - Relative error: -3.565410270607042e-14
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 1296.8699999999997
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.7532495581147846e-16
- (4, 3, 1)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0

- (5, 4, 1)
 - Value: 1422.06
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.598903530394161e-16

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 384.46226625240024
 - Absolute error: 6.048139766789973e-11
 - Relative error: 1.5731426196245508e-13
- (2, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2841.16
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.6005693128386436e-16
- (4, 3, 1)
 - Value: 1857.2400000000002
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.2242557528549464e-16
- (5, 4, 1)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



$$(a) \ t^{(H)} - (1, 2)$$



$$(a) \ t^{(H)} - (2, 1)$$



$$(a) \ t^{(H)} - (2, 1)$$



$$(a) \ t^{(H)} - (3, 1)$$



$$(a) \ t^{(H)} - (4, 1)$$



$$(a) \ t^{(H)} - (5, 1)$$



$$(a) \ t^{(H)} - (1, 1, 2)$$



$$(a) \ t^{(H)} - (2, 1, 1)$$



(a) $t^{(H)} - (2, 2, 1)$



(a) $t^{(H)} = (3, 5, 1)$

(a) $t^{(H)} = (4, 3, 1)$



(a) $t^{(H)} = (5, 4, 1)$

(a) $t^{(C)} = (1, 3)$

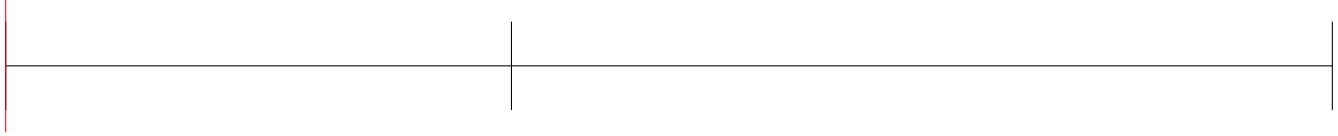


(a) $t^{(C)} = (1, 2)$

(a) $t^{(C)} = (2, 2)$



(a) $t^{(C)} = (5, 2)$



(a) $t^{(C)} - (3, 2)$



(a) $t^{(C)} - (4, 2)$



(a) $t^{(C)} - (1, 1, 2)$



(a) $t^{(C)} - (2, 1, 1)$



(a) $t^{(C)} - (2, 2, 1)$



(a) $t^{(C)} - (3, 5, 1)$



(a) $t^{(C)} - (4, 3, 1)$



(a) $t^{(C)} - (5, 4, 1)$

5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 3.759394
 - Error: 0.000000000002854
 - Relative error: -0.000000000000759
- $(2, 1, 1)$
 - Value: 9.439496
 - Error: 0.000000000002473
 - Relative error: 0.000000000000262
- $(2, 2, 1)$
 - Value: 30.932718
 - Error: 0.000000000000004
 - Relative error: 0.000000000000000
- $(3, 5, 1)$
 - Value: 71.165974
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(4, 3, 1)$
 - Value: 23.670260
 - Error: 0.000000000000011
 - Relative error: -0.000000000000000
- $(5, 4, 1)$
 - Value: 82.719688
 - Error: 0.000000000000014
 - Relative error: -0.000000000000000

5.2 A_{CU}

- $cu, 1$
 - Value: 5.228012
 - Error: 0.000000000000328
 - Relative error: -0.000000000000063
- $cu, 3$
 - Value: 16.738011
 - Error: 0.000000000000327
 - Relative error: 0.000000000000020
- $cu, 5$

- Value: 16.234427
- Error: 0.000000000001808
- Relative error: -0.000000000000111



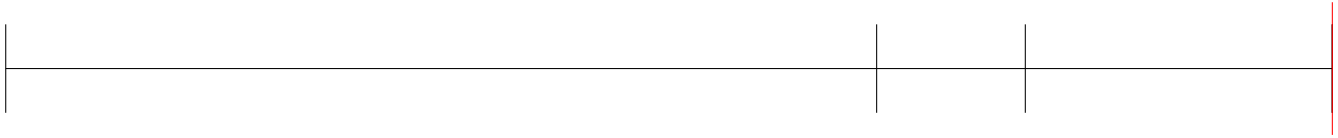
(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (3, 5, 1)$



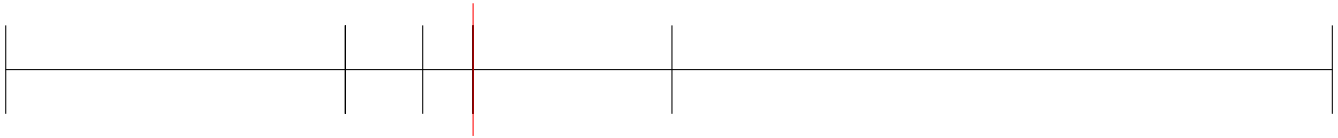
(a) $A^\beta - (4, 3, 1)$



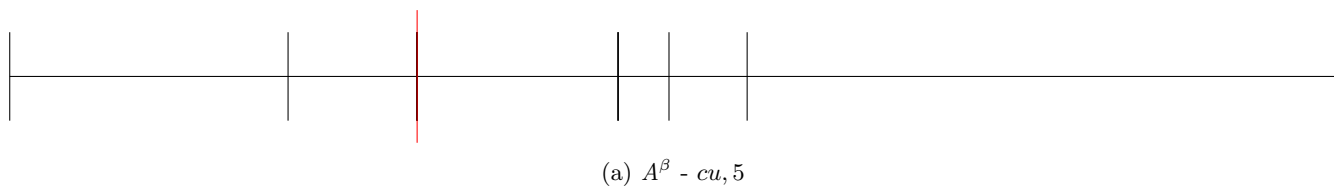
(a) $A^\beta - (5, 4, 1)$



(a) $A^\beta - cu, 1$



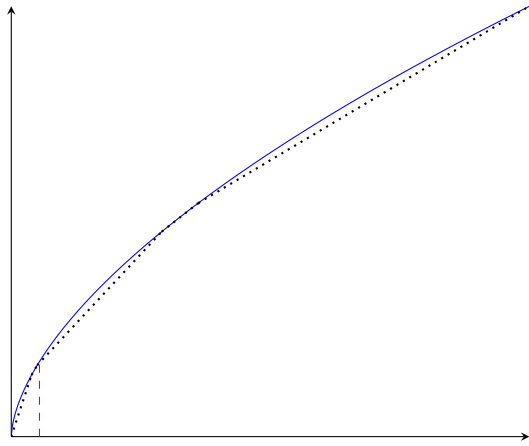
(a) $A^\beta - cu, 3$



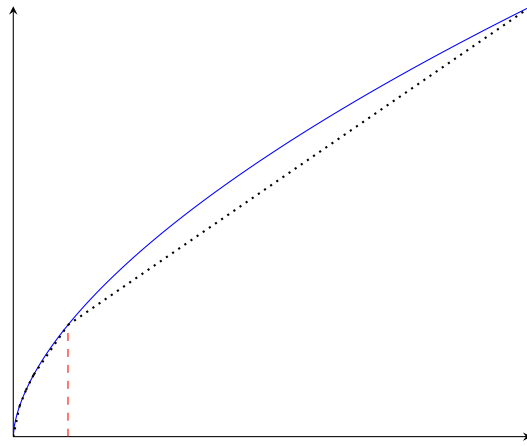
6 A^β Breakpoints

- (1, 1, 2)
 - q : 3.759394
 - $\hat{q}^\beta$: 2.177072
 - q^β : 2.213455
 - Breakpoints: [0, 2.7005882352941173, 3.460657649345137, 20.05363555217427, 24.927399131526517, 24.927400386350968, 24.928068483287923, 24.928068483288385, 69.05411764705883]
 - Error: 0.036382971935908
 - Relative error: 0.016437186075690
- (2, 1, 1)
 - q : 9.439496
 - $\hat{q}^\beta$: 3.845646
 - q^β : 3.845646
 - Breakpoints: [0, 1.0776943916793027, 2.1978599029887658, 3.3036464983028657, 3.4652942623256235, 3.551463972188699, 9.439495705511714, 89.41176470588236]
 - Error: 0.000000000000261
 - Relative error: -0.000000000000068
- (2, 2, 1)
 - q : 30.932718
 - $\hat{q}^\beta$: 7.838822
 - q^β : 7.838823
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 16.51257309583564, 18.676744629514324, 18.890814020083003, 19.107703216652027, 30.93278282491032, 30.932782824910646, 30.93278282491164, 76.14235294117645]
 - Error: 0.000000933342727
 - Relative error: 0.000000119066695
- (3, 5, 1)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.923003
 - q^β : 12.923003
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, 49.559658500720396, 71.16597394333012, 71.16597394333036, 71.16597394333074, 181.68117647058827]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (4, 3, 1)
 - q : 23.670260
 - $\hat{q}^\beta$: 6.676083
 - q^β : 6.676084

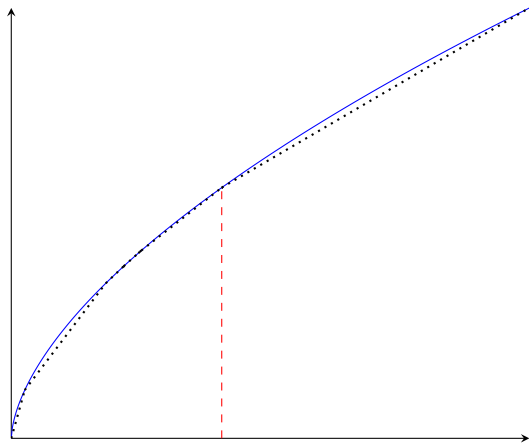
- Breakpoints: [0, 19.921388087224845, 23.670439786681058, 23.670439786681666, 23.670439786681715, 181.14352941176475]
- Error: 0.000001045238617
- Relative error: 0.000000156564622
- (5, 4, 1)
 - q : 82.719688
 - $\hat{q}^\beta$: 14.143775
 - q^β : 14.143775
 - Breakpoints: [0, 7.000222767756085, 12.878068451030797, 38.682379459353704, 65.27329951717095, 82.70899711080247, 82.7196877037195, 192.3352941176471]
 - Error: 0.0000000000000009
 - Relative error: 0.0000000000000001
- $cu, 1$
 - q : 5.228012
 - $\hat{q}^\beta$: 2.682203
 - q^β : 2.697752
 - Breakpoints: [0, 4.5836803758229046, 7.527593053896052, 7.5275930538960525, 7.5275930538960925, 22.75735506822926]
 - Error: 0.015548631793501
 - Relative error: 0.005763551960838
- $cu, 3$
 - q : 16.738011
 - $\hat{q}^\beta$: 5.422784
 - q^β : 5.422784
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.738084386188994, 16.738084386191687, 16.73892021814917, 16.73899708924428, 16.738997089244634, 16.738997089244645, 18.852085704732907, 20.61398815516275, 20.613988155162765, 128.44338870924773]
 - Error: 0.000000428959368
 - Relative error: 0.000000079103161
- $cu, 5$
 - q : 16.234427
 - $\hat{q}^\beta$: 5.324294
 - q^β : 5.324296
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.23492858720698, 16.234939065695293, 16.234975184591587, 16.235135805353295, 20.944873037188618, 20.944873037188774, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.000001978039117
 - Relative error: 0.000000371511876



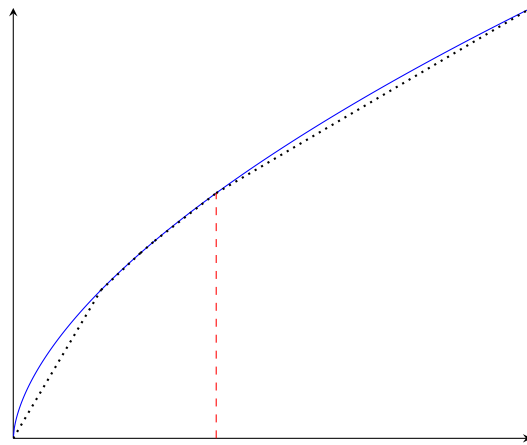
(a) $q^\beta - (1, 1, 2)$



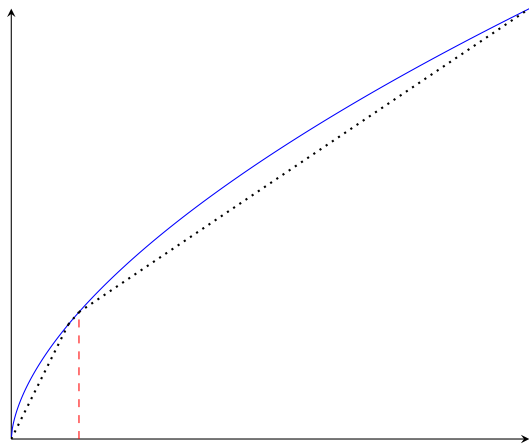
(b) $q^\beta - (2, 1, 1)$



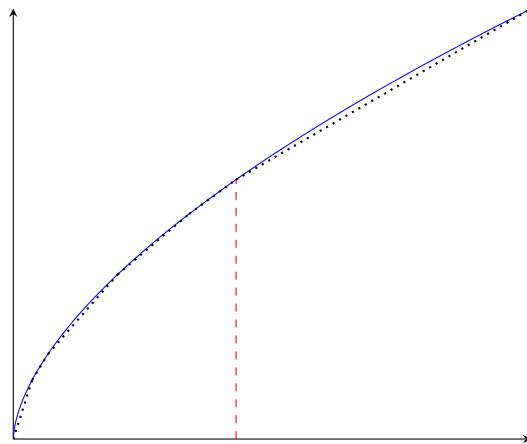
(c) $q^\beta - (2, 2, 1)$



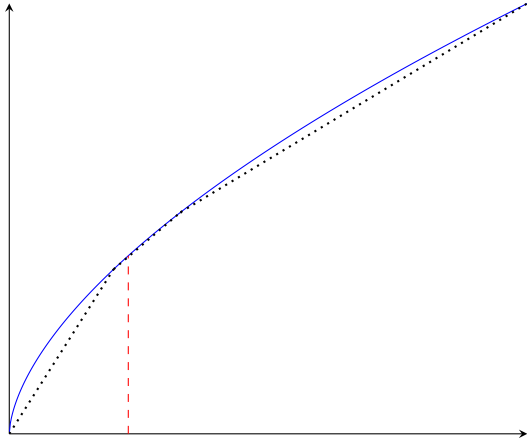
(d) $q^\beta - (3, 5, 1)$



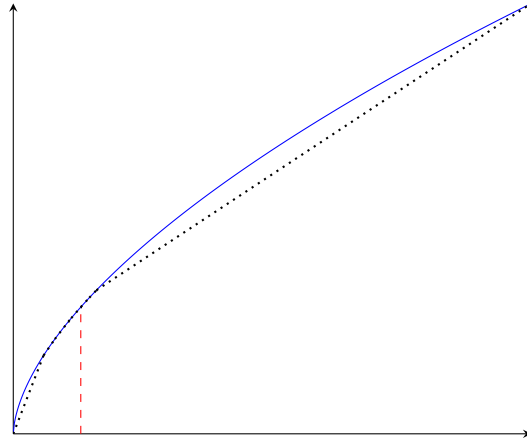
(e) $q^\beta - (4, 3, 1)$



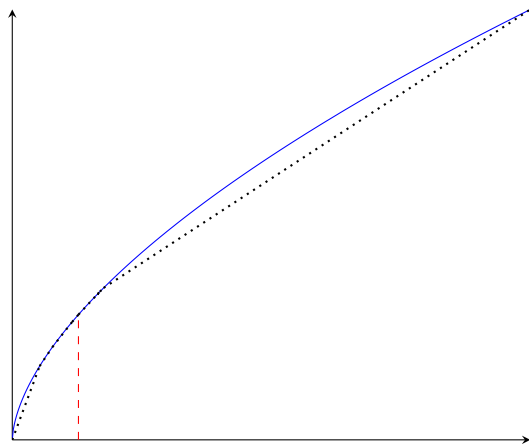
(f) $q^\beta - (5, 4, 1)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 3$



(c) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,1}$	0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{4,5,2}$	0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	0.0	$z_{5,1,1}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,1,2}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,2}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,2,2}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,2}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,3,2}$	0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	0.0	$z_{5,4,2}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,2}$	0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{5,5,2}$	0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{cu,2}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,2}$	0.0	$z_{cu,4}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,1}$	0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,2}$	0.0	$z_{hu,2}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	0.0	$z_{hu,3}$	0.0	$A_{4,3,2}^\beta$	0.0
$z_{2,5,2}$	0.0	$z_{hu,4}$	0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	0.0	$z_{hu,5}$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,2}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,2}$	0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,2,2}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,2}$	0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,3,2}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{2,1,3}$	100.0
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,2,2}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,1}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,2}$	10.0
$A_{1,4,1}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,4,2}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,2}$	10.0
$A_{1,5,2}$	0.0	$A_{5,4,2}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,1,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,2,2}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,3,1}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,3}$	156.75608108107988
$A_{2,3,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,4,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,4,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	111.1
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{3,1,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,2}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,4,2}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,2}$	10.0	$dt_{3,4,3}$	144.5
$A_{3,3,1}$	0.0	$dt_{1,2,3}$	10.0	$dt_{3,5,3}$	133.4
$A_{3,3,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,1,2}$	25.159436961416457
$A_{3,4,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,1,3}$	100.0
$A_{3,5,2}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,4,2}$	10.0	$dt_{4,2,2}$	10.0
$A_{4,1,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,2,3}$	106.1
$A_{4,2,1}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,3}$	183.3
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	10.0	$f_{1,5,2}^H$	0.0	$f_{5,4,2}^H$	0.0
$dt_{4,4,3}$	94.5	$f_{2,1,2}^H$	0.0	$f_{5,5,1}^H$	5.435558889722436
$dt_{4,5,1}$	122.2	$f_{2,2,2}^H$	0.0	$f_{5,5,2}^H$	7.644980900862544
$dt_{4,5,2}$	10.0	$f_{2,3,1}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,2}^H$	10.6	$f_{1,2,1}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,4,1}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	10.0	$f_{2,5,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,2}$	10.0	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,2}^H$	0.0	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	5.214090626939737	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,4,3}$	116.7	$f_{3,3,2}^H$	2.122276361818174	$f_{2,3,1}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,2}^H$	12.677723638181721	$f_{2,4,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{2,5,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	12.6	$f_{3,1,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,1}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,2,2}^H$	5.358470764617547	$f_{5,1,1}^H$	0.0	$f_{3,4,2}^C$	17.3
$f_{1,3,1}^H$	0.0	$f_{5,1,2}^H$	10.055019099137446	$f_{3,5,2}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,1,2}^C$	5.0181288957933265
$f_{1,4,2}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,3,2}^C$	8.4	$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,2,1}$	0.0
$f_{4,4,1}^C$	0.0	$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,2}$	0.0
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,3,1}$	0.0
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,2}$	0.0
$f_{4,5,2}^C$	13.9	$\text{RecLMTD}_{3,4,1}$	0.1	$q_{1,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.1	$q_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.012092773362638837	$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.013380263157893217	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{2,1,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.1	$q_{2,2,2}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,3,1}$	0.0
$f_{5,4,2}^C$	0.0	$\text{RecLMTD}_{4,3,2}$	0.1	$q_{2,3,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.020576623473060753	$q_{2,4,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.022300296262842634	$q_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.1	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.1	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.1	$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.1	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.02880769543069063	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,4,2}$	0.1	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.06524922356449561	$\text{RecLMTD}_{5,5,2}$	0.009469696969696968	$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115768875	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.1	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,3,2}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$q_{1,1,1}$	0.0	$q_{4,5,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{4,5,2}$	0.0	$t_{2,5,2}^H$	198.85754716980975	$t_{1,4,1}^C$	102.8602026715427
$q_{5,1,1}$	0.0	$t_{3,1,1}^H$	105.65126083492889	$t_{1,4,2}^C$	129.3704699676358
$q_{5,1,2}$	0.0	$t_{3,1,2}^H$	65.6	$t_{1,5,1}^C$	204.4
$q_{5,2,1}$	0.0	$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,2}^C$	60.0
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	143.0726937444027	$t_{2,3,1}^C$	37.8
$q_{5,4,2}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	100.93139891203774	$t_{2,4,1}^C$	91.38451456131654
$q_{5,5,2}$	0.0	$t_{3,5,2}^H$	65.6	$t_{2,4,2}^C$	82.2
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,1}^C$	204.4
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	148.9	$t_{2,5,2}^C$	119.0129496402887
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,1}^C$	127.37875345281365
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,1,2}^C$	60.0
$q_{hu,3}$	0.0	$t_{4,3,2}^H$	148.9	$t_{3,2,1}^C$	128.0783721540173
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,3,1}^C$	111.50012781014851
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,2,1}^H$	160.0	$t_{4,5,2}^H$	148.9	$t_{3,4,1}^C$	82.2
$t_{1,2,2}^H$	160.0	$t_{5,1,1}^H$	179.34745762711862	$t_{3,4,2}^C$	82.2
$t_{1,3,1}^H$	160.0	$t_{5,1,2}^H$	80.13309079349328	$t_{3,5,2}^C$	93.3
$t_{1,3,2}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	65.88648737862955
$t_{1,4,1}^H$	160.0	$t_{5,2,2}^H$	65.6	$t_{4,1,2}^C$	60.0
$t_{1,4,2}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	138.76511970103755
$t_{1,5,1}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,2,2}^C$	115.6
$t_{1,5,2}^H$	160.0	$t_{5,4,2}^H$	65.6	$t_{4,3,2}^C$	37.8
$t_{2,1,2}^H$	233.60621120462815	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	82.2
$t_{2,2,2}^H$	187.84245283018862	$t_{5,5,2}^H$	92.0025024268445	$t_{4,4,2}^C$	82.2
$t_{2,3,1}^H$	248.9	$t_{1,1,1}^C$	160.0	$t_{4,5,1}^C$	204.4
$t_{2,3,2}^H$	137.8	$t_{1,2,1}^C$	176.10015191639366	$t_{4,5,2}^C$	93.3
$t_{2,4,1}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,1}^C$	92.62147189941555
$t_{2,4,2}^H$	137.8	$t_{1,3,1}^C$	111.5	$t_{5,1,2}^C$	60.0
$t_{2,5,1}^H$	148.81509433962145	$t_{1,3,2}^C$	215.35238095238094	$t_{5,2,1}^C$	176.10015191639366

Variable	Value	Variable	Value	Variable	Value
$t_{5,2,2}^C$	115.6	$b_{4,1,2}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0
$t_{5,3,1}^C$	221.1	$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0
$t_{5,3,2}^C$	37.8	$b_{4,2,2}^{H,in}$	1876.14	$b_{3,1,2}^{H,out}$	0.0
$t_{5,4,2}^C$	146.03815028901732	$b_{4,3,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1182.0343451272383
$t_{5,5,1}^C$	204.4	$b_{4,4,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0
$t_{5,5,2}^C$	93.3	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	456.3066254370861
$b_{1,2,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	857.3553223388075	$b_{5,1,1}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	1354.5633745628793
$b_{1,3,1}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	916.844959948186	$b_{3,5,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	1876.14
$b_{1,5,2}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	0.0	$b_{4,3,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	1081.1326631657926	$b_{4,4,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	968.8350400517452	$b_{4,4,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	1460.6800000000012	$b_{1,2,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	857.3553223388074	$b_{5,1,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	916.844959948186
$b_{2,5,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	1182.0343451272383	$b_{1,5,2}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	1081.1326631657926
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	968.8350400517452
$b_{3,3,2}^{H,in}$	456.3066254370861	$b_{2,3,1}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	1460.6800000000012	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	1354.5633745628793	$b_{2,4,1}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{1,4,1}^{C,in}$	0.0	$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	301.0877337475996
$b_{1,4,2}^{C,in}$	0.0	$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0
$b_{1,5,1}^{C,in}$	0.0	$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0
$b_{1,5,2}^{C,in}$	0.0	$b_{5,4,2}^{C,in}$	0.0	$b_{4,3,2}^{C,out}$	317.52
$b_{2,1,2}^{C,in}$	0.0	$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0
$b_{2,2,2}^{C,in}$	705.16	$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{2,3,1}^{C,in}$	0.0	$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	1296.87
$b_{2,4,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,4,2}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	705.16	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	1422.0600000000002	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	301.0877337475996	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,3,2}^{C,in}$	317.52	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	1296.87	$b_{3,4,1}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	1422.0600000000002		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		